

DEVESH RAVICHANDRAN

AI & Machine Learning Engineer | Generative AI | RAG | Python

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EDUCATION

SRM Institute of Science and Technology Chennai, India | 2022 – 2026

B.Tech — Computer Science, Specialisation in Artificial Intelligence & Machine Learning | **CGPA: 7.2**

- Worked on an AI Assistant for Dean (In-House Project)
- Participated in 2+ national-level hackathons representing SRM

PROFESSIONAL EXPERIENCE

DevOps Intern | CaratLane — A Tanishq Partnership Jan 2026 – Apr 2026

- Evaluated and implemented an AWS Client VPN proof of concept for secure access to internal resources.
- Developed a User Access Review (UAR) tool to automate periodic access-audit workflows across internal systems.
- Automated Freshservice ticket workflows using scripting and integration pipelines to reduce repetitive manual operations.
- Worked with CI/CD pipelines, Docker, AWS infrastructure, and production support workflows.

AI/ML Intern | ONGC Dec 2024 – Jan 2025

- Built a RAG-based PDF summarisation chatbot and intelligent query bot for internal document retrieval.
- Integrated an LLM for contextual summarisation, improving document processing speed by 60% and query response by 95%.
- Engineered a Streamlit interface with text cleaning, error handling, and FAISS-based semantic search.

PROJECTS

Loan Portfolio Risk Monitoring Agent

Technologies: Python, RAG, FAISS, LLaMA 3 8B, Agentic Search, DuckDuckGo Web Search

- Built an agentic risk-monitoring system for loan portfolios combining internal Credit Memo information with live external news.
- Implemented RAG-based retrieval using FAISS and LLaMA 3 8B to analyse borrower information and relevant risk signals.
- Integrated web search and designed the workflow to compare retrieved evidence against the Credit Memo before producing a risk decision.

3D Virtual Try-On Clothing Application

Technologies: Python, OpenCV, TensorFlow, Unity 3D, ARCore, React Native

- Built an AI-powered mobile application for virtual 3D clothing try-ons, addressing realistic garment warping across varied body poses.
- Used pose estimation for body mapping and thin-plate spline (TPS) transformations for garment fitting.
- Integrated AR overlays for real-time garment visualisation and tested the computer vision pipeline locally as part of a clothing brand website.

Attention-Guided Multi-Scale Feature Fusion for Skin Disease Classification

Technologies: Python, TensorFlow, Keras, Grad-CAM, LIME

- Designed a CNN with attention gates for dermatological image classification, focusing on diagnostically important image regions.
- Integrated Grad-CAM and LIME for visual interpretability and model explanation.
- Achieved 91% classification accuracy through hyperparameter tuning and augmentation on limited medical datasets.

Frame Interpolation with Explainable AI

Technologies: Python, Deep Learning, XAI Techniques

- Developed a video frame interpolation system that generates intermediate frames from sparse inputs.
- Applied explainability techniques to interpret temporal features influencing interpolation decisions.

PUBLICATION

Comparative Analysis on BERT Models: ALBERT and RoBERTa

Tashin Khan, Thejal Ravichandran, Devesh Ravichandran

TECHNICAL SKILLS

ML / AI: Machine Learning, Deep Learning, Computer Vision, NLP, Generative AI, Agentic AI, RAG, LLMs, Explainable AI, Transformers

Frameworks: TensorFlow, Keras, LangChain, LangGraph, Streamlit, Flask, FAISS, React.js, Express.js

Languages: Python, SQL, JavaScript

Databases: MySQL, PostgreSQL, NoSQL, Elasticsearch

Cloud / DevOps: AWS, Oracle Cloud, Docker, CI/CD pipelines

Tools: Ollama, Gemini API, Meta-LLaMA 3.1, OpenCV, ARCore, Unity 3D

CERTIFICATIONS

- Oracle Cloud Infrastructure: Generative AI Professional
- AWS Academy Machine Learning Foundations
- AWS Academy Cloud Foundations
- AWS Academy Computer Networks

ACHIEVEMENTS

- Founder of a self-funded clothing brand — built an AI-powered virtual try-on system as part of the brand's website.
- Secured 3rd Place in a National-Level Hackathon.
- Mridangam — awarded in multiple elocution competitions.